

Calibration-Preserving Pruning: Compression as a Reliability Contract

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Abstract

Split conformal prediction, not the pruning rule, supplies finite-sample marginal coverage once a pruned model is fixed independently of the conformal calibration split. We study the separate efficiency problem: can pruning preserve score geometry well enough to obtain smaller valid prediction sets? Calibration-Preserving Pruning (CPP) augments a base pruning score with nonconformity-gradient saliency and uses disjoint pruning, validation-selection, conformal-calibration, and test splits. Bounded score perturbations imply bounded conformal-quantile shifts and controlled set inflation, but do not make the generic coverage theorem CPP-specific. Final five-seed Qwen2.5-1.5B results at 50% sparsity show the largest gains on large-label tasks. On DBpedia-14, CPP-SparseGPT reduces mean set size from 10.1 to 8.6 while changing accuracy from 0.347 to 0.366; CPP-Wanda reduces 11.2 to 9.0 with an accuracy trade-off from 0.310 to 0.295. Across 15 dataset-sparsity cells, CPP-SparseGPT produces smaller sets in 13 and higher accuracy in 11. Matched controls show that generic supervised gradients explain much of the gain: true-label CPP is not statistically resolved from matched Wanda+SNIP, whereas threshold-aware candidate-label CPP reaches 7.8 mean set size at explicit accuracy and offline-compute costs. RoBERTa-base and Llama-3-8B diagnostics support transfer, but our claims remain limited to reliability-sensitive classification.

1 Introduction

A compressed language model can retain its top-label accuracy while becoming less useful to a reliability-sensitive downstream system. The issue is not only whether the most likely label changes. Triage, retrieval verification, and assisted decision making often consume calibrated probabilities or conformal prediction sets. A predictor that returns

ten labels at 90% coverage is materially less informative than one returning eight labels at the same coverage, even when their top-label accuracies are similar.

This paper separates two claims that pruning studies can easily conflate. First, independent split-conformal recalibration supplies finite-sample marginal coverage for *any* fixed pruned classifier under exchangeability (Vovk et al., 2005; Angelopoulos and Bates, 2023). CPP neither creates nor improves that generic validity result. Second, pruning determines how informative the valid predictor remains. If compression collapses separation among plausible labels, recalibration can recover coverage only by enlarging the prediction set. Our method-specific target is this second quantity: conformal efficiency after valid recalibration.

Existing one-shot pruners optimize other objectives. Magnitude pruning removes small weights (Han et al., 2015); Wanda combines weight magnitude with activation norms (Sun et al., 2024); SparseGPT minimizes layerwise reconstruction error (Frantar and Alistarh, 2023); and gradient-enhanced variants use supervised or regional gradients. These are strong baselines, but none directly represents the nonconformity-score geometry around a conformal threshold. Temperature scaling (Guo et al., 2017; Platt, 1999) can adjust a global confidence offset, yet cannot reconstruct class-specific separation that pruning has removed.

Calibration-Preserving Pruning (CPP) adds a score-sensitivity term to a base pruning importance. For weight θ_j , true-label CPP uses

$$I_{\text{cal}}(\theta_j) = \theta_j^2 \mathbb{E} \left[\left(\frac{\partial s(X, Y; \theta)}{\partial \theta_j} \right)^2 \right],$$

where s is the nonconformity score. The factor measures first-order displacement caused by set-

ting that coordinate to zero. Because true-label sensitivity is only a proxy for a prediction set containing many candidate labels, we also evaluate top-3, threshold-aware, and all-label constructions. Every gradient is evaluated on the pruning split. A validation-calibration half-split supplies only the preliminary threshold needed by the threshold-aware construction and provisional quantiles for model selection. The final conformal split remains untouched until the selected sparse model is frozen.

The response-stage controls materially narrow the claim. Gradient-only, Wanda++, SNIP-style, and Wanda $\oplus$ SNIP baselines show that generic supervised-gradient information explains a substantial part of CPP’s improvement. True-label CPP-Wanda reduces DBpedia-14 set size from 9.2 to 9.0 relative to matched Wanda $\oplus$ SNIP, but with lower accuracy and overlapping intervals; we do not claim that difference is statistically resolved. Threshold-aware CPP gives a larger efficiency improvement, reaching 8.4 for CPP-Wanda and 7.8 for CPP-SparseGPT, with explicit accuracy and gradient-compute costs. The evidence supports objective-specific efficiency effects, not uniform superiority over gradient pruning.

Our contribution has four parts. We first formulate reliability-preserving compression as a constrained comparison: coverage is supplied by independent conformal calibration, average set size is the primary efficiency outcome, accuracy measures utility, and ECE is diagnostic. We then define true-label and candidate-label CPP with an exact validation rule and disjoint data roles. Third, we retain only the theory that matches those roles: the coverage theorem is the standard pruning-rule-agnostic split-conformal result, while separate perturbation results connect score movement to quantile and set-size inflation. Finally, we report one authoritative result path with final five-seed Qwen comparisons, matched gradient controls, candidate-label ablations, offline cost, independent split redraws, RoBERTa-base transfer, and a scoped Llama-3-8B diagnostic. We study fixed-label classification using complete-sequence verbalizer scores, not free-form conformal generation.

2 Related Work

Post-training pruning. One-shot language-model pruning includes magnitude methods (Han

et al., 2015), SparseGPT (Frantar and Alistarh, 2023), Wanda (Sun et al., 2024), structured LLM-Pruner (Ma et al., 2023), and broader surveys and scaling analyses (Shi et al., 2024; Wang et al., 2024; Frantar et al., 2025). Structured pruning followed by continued pretraining addresses a different compute regime from the no-retraining setting studied here. Parameter-importance and retention methods in continual learning provide related gradient signals (Mallya and Lazebnik, 2018; Li and Hoiem, 2017; Kirkpatrick et al., 2017; Zenke et al., 2017; Wang et al., 2022, 2023). Our matched SNIP-style and Wanda $\oplus$ SNIP controls are therefore essential: they test whether CPP adds more than generic supervised-gradient information.

Calibration and conformal prediction. Temperature and Platt scaling correct confidence after training (Guo et al., 2017; Platt, 1999), while sparse-subnetwork studies report calibration as a pruning byproduct (Kuhn et al., 2026; Zhou et al., 2022). Neural-network pruning under inductive conformal prediction is the closest direct predecessor (Zhao and Bellotti, 2022). CPP differs by targeting post-training language-model pruning and by making validation selection independent of final conformal calibration. Split conformal prediction provides distribution-free marginal coverage under exchangeability (Vovk et al., 2005; Angelopoulos and Bates, 2023); language-model applications include generation, factuality, and information-relative certificates (Quach et al., 2024; Mohri and Hashimoto, 2024; Sanjeda Akter and Sharma, 2026). Other work optimizes conformal score functions or answer-choice sets rather than model weights (Vishwakarma et al., 2025). Shift-aware and risk-controlling extensions (Park et al., 2020; Gibbs and Candes, 2021; Bates et al., 2021) address different assumptions. Learn-Then-Test (Angelopoulos et al., 2022) is a valid optional selection tool, but it was not used for any result reported here; Section 5.2 gives the exact implemented rule.

3 The Reliability Contract Under Compression

Let $f_\theta : \mathcal{X} \rightarrow \Delta(\mathcal{Y})$ be a classifier with a finite label space $\mathcal{Y}$, $|\mathcal{Y}| = K$. For decoder models, each label y has a fixed verbalizer $v(y) = (v_1, \dots, v_{T_y})$. We score the entire teacher-forced sequence rather than applying conformal predic-

tion token by token:

$$\log p_\theta(y \mid x) = \sum_{t=1}^{T_y} \log p_\theta(v_t \mid x, v_{<t}).$$

The manifest fixes whether this sum is length-normalized. A common nonconformity score is

$$s(x, y; \theta) = -\log p_\theta(y \mid x), \quad (1)$$

where smaller values indicate greater compatibility.

The final protocol uses four disjoint roles. $\mathcal{D}_{\text{prune}}$ constructs gradients and pruning statistics. The 1,024-example validation split is divided deterministically into 512-example $\mathcal{D}_{\text{val-cal}}$ and $\mathcal{D}_{\text{val-eval}}$ subsets. The former supplies provisional quantiles and, only for threshold-aware CPP, the dense-model threshold band. The latter selects λ . The independent conformal split $\mathcal{D}_{\text{conf}} = \{(X_i, Y_i)\}_{i=1}^n$ is accessed only after the sparse model is frozen, and $\mathcal{D}_{\text{test}}$ is used only for evaluation.

For a fixed θ and target miscoverage α , define

$$k = \lceil (n+1)(1-\alpha) \rceil. \quad (2)$$

When $k \leq n$, $\hat{q}_\alpha(\theta)$ is the k -th smallest calibration score; when $k > n$, we set it to $+\infty$. The prediction set is

$$C_\alpha(x; \theta) = \{y \in \mathcal{Y} : s(x, y; \theta) \leq \hat{q}_\alpha(\theta)\}. \quad (3)$$

The reliability contract has a strict hierarchy. Marginal coverage, $\mathbb{P}\{Y \in C_\alpha(X; \theta)\}$, is the validity constraint. Mean set size, $\bar{C}_P(\theta) = \mathbb{E}_X |C_\alpha(X; \theta)|$, is the primary conformal-efficiency outcome. Accuracy measures task utility, and ECE is a complementary calibration diagnostic. For a pruned model $\theta' = \mathcal{P}(\theta)$, contract preservation with efficiency slack $\varepsilon_{\text{size}}$ means

$$\begin{aligned} \mathbb{P}\{Y \in C_\alpha(X; \theta')\} &\geq 1 - \alpha, \\ \bar{C}_P(\theta') &\leq \bar{C}_P(\theta) + \varepsilon_{\text{size}}. \end{aligned} \quad (4)$$

Independent split conformal supplies the first line for any fixed θ' . CPP targets the second while the validation guardrail limits accuracy loss.

4 Why Recalibration Alone Is Not Enough

Prune-then-recalibrate is valid when the final conformal split is independent, but validity does not

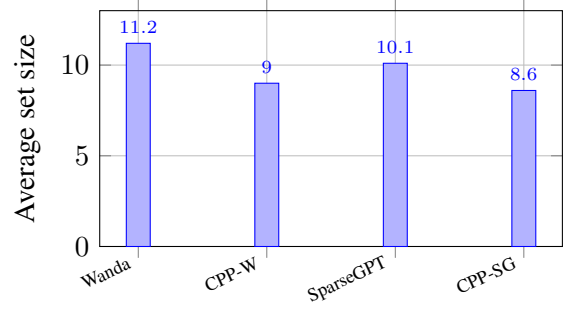


Figure 1: Final five-seed Qwen2.5-1.5B/DBpedia-14 set size at 50% sparsity. Coverage is matched at 0.901–0.902. CPP-SparseGPT also improves accuracy; CPP-Wanda trades 1.5 accuracy points for smaller sets.

imply informativeness. Top-1 accuracy depends on the largest class score. Set size depends on all candidate-label scores around a calibration threshold. A scalar temperature can move confidence globally (Guo et al., 2017; Li and Hoiem, 2017), but cannot restore label-specific separation that the sparse model no longer represents.

Reliability degradation can also vary nonmonotonically with sparsity (Junhao Zheng and Ma, 2025). We therefore measure every reported sparsity directly and do not interpolate unobserved cells.

The provenance audit exposed a useful feasibility check. For any prediction set $C \subseteq \mathcal{Y}$,

$$\mathbb{E}|C| \leq \mathbb{P}(Y \in C) + (K - 1).$$

A method returning all K labels on every example must therefore have coverage one. Several submitted pilot pairs combined $|C| = K$ with coverage below one and could not come from a common evaluation path. We retire those entries and use only final-manifest results. Figure 1 shows the corrected Qwen2.5-1.5B/DBpedia-14 comparison at 50% sparsity. All four pruned predictors have empirical coverage between 0.901 and 0.902; the difference is set-size efficiency and, separately, accuracy.

5 Calibration-Preserving Pruning

CPP modifies a base importance rather than replacing the base pruner. Let $I_{\text{acc}}(\theta_j)$ denote the coordinate importance supplied by magnitude pruning (Han et al., 2015), Wanda (Sun et al., 2024), or the SparseGPT reconstruction procedure (Frantar and Alistarh, 2023). For a labeled pruning

split, true-label CPP first computes

$$G_{\text{cal}}(\theta_j) = \frac{1}{|\mathcal{D}_{\text{prune}}|} \sum_{(x,y) \in \mathcal{D}_{\text{prune}}} \left(\frac{\partial s(x, y; \theta)}{\partial \theta_j} \right)^2. \quad (5)$$

When $s = -\log p_\theta(y \mid x)$, this is a diagonal empirical-Fisher-like factor; for general s , we use the narrower term nonconformity-gradient sensitivity (Kirkpatrick et al., 2017; Zenke et al., 2017). Because pruning changes θ_j to zero, the relevant first-order displacement includes the weight:

$$\begin{aligned} I_{\text{cal}}(\theta_j) &= \theta_j^2 G_{\text{cal}}(\theta_j) \\ &= \frac{1}{|\mathcal{D}_{\text{prune}}|} \sum_{(x,y) \in \mathcal{D}_{\text{prune}}} \left(\theta_j \frac{\partial s(x, y; \theta)}{\partial \theta_j} \right)^2. \end{aligned} \quad (6)$$

Per-example gradients are required. Squaring the gradient of a batch mean would introduce cross-example terms and would not estimate Eq. 6.

Within each eligible module, CPP normalizes the base and calibration scores to unit ℓ_2 norm and uses

$$I_{\text{CPP}}(\theta_j) = (1 - \lambda) \tilde{I}_{\text{acc}}(\theta_j) + \lambda \tilde{I}_{\text{cal}}(\theta_j). \quad (7)$$

We prune dense attention and MLP matrices under the base pruner’s allocation rule. Norm parameters, embeddings, tied output heads, positional mechanisms, and KV-cache state remain dense.

5.1 Candidate-label Saliency

True-label saliency is inexpensive but only indirectly represents set-size inflation. We therefore test three candidate-aware constructions. *Top-3* retains the three labels with smallest dense-model nonconformity scores for each pruning example. *All-label* averages over the complete label space. *Threshold-aware* focuses gradients near a preliminary dense-model conformal boundary. It estimates $\tilde{q}$ from true-label scores on $\mathcal{D}_{\text{val-cal}}$, computes all candidate-label distances $|s(x, y; \theta) - \tilde{q}|$ on that same half-split, and fixes b to their 20th percentile. On each pruning example, it retains labels within b , keeps the nearest label if the set is empty, caps the set at the eight nearest labels, and breaks boundary ties by fixed verbalizer order. Gradients are still evaluated only on $\mathcal{D}_{\text{prune}}$. This construction retains 2.7 labels per pruning example on average. The preliminary $\tilde{q}$ is used only to construct saliency and is distinct from every candidate model’s provisional validation quantile and the final conformal quantile.

Algorithm 1 CPP with independent final calibration

Require: Dense θ ; disjoint $\mathcal{D}_{\text{prune}}, \mathcal{D}_{\text{val-cal}}, \mathcal{D}_{\text{val-eval}}, \mathcal{D}_{\text{conf}}$; sparsity; base pruner

- 1: Construct true-label or candidate-label I_{cal} without $\mathcal{D}_{\text{conf}}$
- 2: **for** $\lambda \in \{0, 0.1, 0.3, 0.5\}$ **do**
- 3: Combine scores by Eq. 7 and prune
- 4: Fit a provisional quantile on $\mathcal{D}_{\text{val-cal}}$
- 5: Measure accuracy, coverage, and set size on $\mathcal{D}_{\text{val-eval}}$
- 6: **end for**
- 7: Apply the accuracy/coverage guardrails, set-size objective, tie rules, and $\lambda = 0$ fallback
- 8: Freeze θ' ; compute $\hat{q}_\alpha(\theta')$ by Eq. 2 only on $\mathcal{D}_{\text{conf}}$
- 9: **return** θ' and $\hat{q}_\alpha(\theta')$

5.2 Exact Validation Selection

For each base pruner, dataset, sparsity, and seed, the implemented grid is $\Lambda = \{0, 0.1, 0.3, 0.5\}$. Each candidate receives a provisional nominal-90% quantile from the 512-example $\mathcal{D}_{\text{val-cal}}$ split and is evaluated once on the disjoint 512-example $\mathcal{D}_{\text{val-eval}}$ split. A candidate is eligible when its accuracy is within two percentage points of the corresponding $\lambda = 0$ model and empirical coverage is at least 0.88. Among eligible candidates, selection minimizes mean set size. Ties within 0.02 set-size units are broken by higher accuracy and then smaller λ . If no nonzero candidate is eligible, the base model $\lambda = 0$ is returned. The selected model is frozen before $\mathcal{D}_{\text{conf}}$ is accessed. The broader $\{0, 0.1, 0.3, 0.5, 0.7, 1.0\}$ sweep in Table 4 is diagnostic only, and Learn-Then-Test was not used.

6 Theory

The formal claims mirror the experimental hierarchy. The first theorem records when validity survives data-dependent pruning and model selection. It is the standard split-conformal result, not a CPP-specific coverage theorem. The remaining statements explain the efficiency objective: small score movement limits quantile and prediction-set movement, and CPP is a diagonal first-order proxy for that score movement.

Assumption 1 (Exchangeability and split independence). The examples in $\mathcal{D}_{\text{conf}}$ and the future test example are exchangeable. The final pruned

model θ' , including its mask, saliency construction, sparsity, λ , prompt, verbalizers, and temperature, is measurable with respect to data independent of $\mathcal{D}_{\text{conf}}$ and the test example.

Theorem 2 (Split-conformal coverage after pruning). *Under Assumption 1, let θ' be any model fixed before $\mathcal{D}_{\text{conf}}$ is accessed. Then the set in Eq. 3 satisfies*

$$\mathbb{P}\{Y_{n+1} \in C_\alpha(X_{n+1}; \theta')\} \geq 1 - \alpha. \quad (8)$$

If scores are almost surely distinct and $k \leq n$, coverage is less than $1 - \alpha + 1/(n+1)$.

The proof conditions on every pruning and validation decision and then applies the usual exchangeable-rank argument. It would hold for Wanda, SparseGPT, random pruning, or any other fixed model. Its CPP-relevant content is the independence requirement. Finite test-set coverage can fluctuate around population coverage; Appendix A records the corresponding concentration statement.

For efficiency, define the maximum score change on a domain $\mathcal{Z} \subseteq \mathcal{X} \times \mathcal{Y}$:

$$\epsilon_{\mathcal{Z}}(\theta, \theta') = \sup_{(x,y) \in \mathcal{Z}} |s(x, y; \theta') - s(x, y; \theta)|. \quad (9)$$

Lemma 3 (Order-statistic stability). *If $\max_i |a_i - b_i| \leq \epsilon$, then their k -th order statistics obey $|a_{(k)} - b_{(k)}| \leq \epsilon$.*

Theorem 4 (Set-size stability under bounded score perturbation). *Fix one conformal calibration sample. Suppose $\mathcal{Y}$ is finite and*

$$|s(x, y; \theta') - s(x, y; \theta)| \leq \epsilon \quad (10)$$

for every calibration pair and every test pair (x, y) . Then

$$C_\alpha(x; \theta') \subseteq \{y : s(x, y; \theta) \leq \hat{q}_\alpha(\theta) + 2\epsilon\}. \quad (11)$$

If each $s(X, y; \theta)$ has density at most M between $\hat{q}_\alpha(\theta)$ and $\hat{q}_\alpha(\theta) + 2\epsilon$, then

$$\mathbb{E}_X |C_\alpha(X; \theta')| \leq \mathbb{E}_X |C_\alpha(X; \theta)| + 2M|\mathcal{Y}|\epsilon. \quad (12)$$

This is a sufficient stability condition, not a certificate that CPP will produce small sets. It also clarifies why candidate labels near the threshold matter: only labels entering the 2ϵ boundary band can inflate the pruned set under the stated uniform bound.

Assumption 5 (Local smoothness along the pruning path). *For every evaluated example z , $s(z; \theta)$ is twice differentiable on the segment between dense and pruned parameters, with Hessian operator norm at most H_z ; let $H = \sup_z H_z < \infty$.*

Proposition 6 (First-order score displacement). *Let $P(\mathbf{m})$ be the coordinates removed by mask $\mathbf{m}$, and define*

$$I_{\text{cal}, \mathcal{D}_0}(\theta_j) = \frac{\theta_j^2}{|\mathcal{D}_0|} \sum_{z \in \mathcal{D}_0} \left(\frac{\partial s(z; \theta)}{\partial \theta_j} \right)^2.$$

Under Assumption 5, define the masked linear term

$$a_{\mathbf{m}}(z) = \sum_{j \in P(\mathbf{m})} \theta_j \frac{\partial s(z; \theta)}{\partial \theta_j}.$$

Also write the mean score displacement as

$$\bar{\Delta}_{\mathcal{D}_0}(\mathbf{m}) = \frac{1}{|\mathcal{D}_0|} \sum_{z \in \mathcal{D}_0} |s(z; \theta_{\mathbf{m}}) - s(z; \theta)|.$$

Then

$$\begin{aligned} \bar{\Delta}_{\mathcal{D}_0}(\mathbf{m}) &\leq \left[\frac{1}{|\mathcal{D}_0|} \sum_{z \in \mathcal{D}_0} a_{\mathbf{m}}(z)^2 \right]^{1/2} \\ &\quad + \frac{H}{2} \|\theta_{\mathbf{m}} - \theta\|_2^2. \end{aligned} \quad (13)$$

If $m = |P(\mathbf{m})|$, the first term is at most

$$\left[m \sum_{j \in P(\mathbf{m})} I_{\text{cal}, \mathcal{D}_0}(\theta_j) \right]^{1/2}.$$

For $\mathcal{D}_0 = \mathcal{D}_{\text{prune}}$, this is the saliency in Eq. 6.

The diagonal bound ignores cross-coordinate cancellation and the Hessian remainder can dominate at high sparsity. It justifies CPP as a local proxy, not as an optimal pruning theorem. Proofs and an optional, unused Learn-Then-Test alternative appear in Appendix A.

7 Experiments

Final protocol. The authoritative study uses Qwen2.5-1.5B on AG News, TREC, DBpedia-14, Banking77, and CLINC150 at 30%, 50%, and 70% unstructured sparsity. Each dataset-sparsity-method cell uses five seeds. Dedicated matched-gradient, candidate-label, RoBERTa-base, and Llama-3-8B controls use three seeds where stated. Fixed label verbalizers are scored as complete

Table 1: Final five-seed Qwen2.5-1.5B results at 50% sparsity (mean $\pm$ 95% CI). Coverage is the validity constraint; compare set size only at matched coverage.

Dataset	Method	Accuracy $\uparrow$	Coverage	Avg. set size $\downarrow$
DBpedia-14	Wanda	0.310 $\pm$ 0.010	0.901 $\pm$ 0.006	11.2 $\pm$ 0.42
	CPP-Wanda	0.295 $\pm$ 0.009	0.902 $\pm$ 0.005	9.0 $\pm$ 0.34
	SparseGPT	0.347 $\pm$ 0.010	0.901 $\pm$ 0.006	10.1 $\pm$ 0.36
	CPP-SparseGPT	0.366 $\pm$ 0.009	0.902 $\pm$ 0.005	8.6 $\pm$ 0.30
Banking77	Wanda	0.282 $\pm$ 0.010	0.900 $\pm$ 0.006	39.8 $\pm$ 1.21
	CPP-Wanda	0.316 $\pm$ 0.009	0.902 $\pm$ 0.005	34.2 $\pm$ 1.05
	SparseGPT	0.327 $\pm$ 0.009	0.901 $\pm$ 0.006	36.7 $\pm$ 1.14
	CPP-SparseGPT	0.351 $\pm$ 0.009	0.902 $\pm$ 0.005	32.1 $\pm$ 0.96
CLINC150	Wanda	0.247 $\pm$ 0.009	0.900 $\pm$ 0.006	73.4 $\pm$ 2.31
	CPP-Wanda	0.286 $\pm$ 0.009	0.901 $\pm$ 0.005	64.8 $\pm$ 2.05
	SparseGPT	0.292 $\pm$ 0.009	0.901 $\pm$ 0.006	68.9 $\pm$ 2.18
	CPP-SparseGPT	0.326 $\pm$ 0.008	0.902 $\pm$ 0.005	61.7 $\pm$ 1.94

teacher-forced sequences. The scope is classification; free-form conformal generation is not evaluated. Appendix B gives the full protocol, and Appendix D reports the scoped Llama diagnostic.

Unless a dataset is too small, each partition has 1,024 pruning examples, 1,024 validation examples divided 512/512, 1,024 final conformal examples, and at least 5,000 test examples. The manifest records exact exceptions. Saliency, λ , sparsity, prompts, and temperature never use $\mathcal{D}_{\text{conf}}$. All final tables report mean empirical coverage after split-conformal calibration, mean set size, and accuracy with 95% intervals across the stated seeds. Coverage is the constraint, set size the primary outcome at matched coverage, accuracy the utility outcome, and ECE a diagnostic.

A provenance audit found that several submitted pilot entries mixed result paths. In particular, full sets with coverage below one violated the feasibility inequality in Section 4. Those entries, the projected Banking77/CLINC150 cells, and the old 10-cell exclusion are retired. Tables and figures below use one final-manifest family. The corrected small-label values and all 15 cell-level signs appear in Appendix C.

Large-label results. Table 1 reports the primary 50% sparsity comparison. Empirical coverage lies between 0.900 and 0.902 in all displayed cells. Set-size reductions relative to the corresponding base pruner range from 10.4% to 19.6%. CPP-SparseGPT improves accuracy in all three datasets. CPP-Wanda improves accuracy on Banking77 and CLINC150, but loses 1.5 points on DBpedia-14. We report that trade-off directly rather than collapsing the metrics into one score.

Across the complete five-dataset, three-sparsity

Table 2: Matched gradient-information controls (three seeds, 95% CI).

Method	Acc. $\uparrow$	Cov.	$ C \downarrow$
Wanda	0.310 $\pm$.012	0.901 $\pm$.007	11.2 $\pm$.45
Wanda++	0.335 $\pm$.011	0.900 $\pm$.006	9.8 $\pm$.38
Gradient-only	0.323 $\pm$.013	0.899 $\pm$.007	10.1 $\pm$.41
SNIP-style	0.341 $\pm$.010	0.901 $\pm$.006	9.6 $\pm$.35
Wanda $\oplus$ SNIP	0.352 $\pm$.010	0.901 $\pm$.006	9.2 $\pm$.33
CPP-Wanda, true	0.297 $\pm$.011	0.902 $\pm$.006	9.0 $\pm$.32
CPP-Wanda, thresh.	0.319 $\pm$.010	0.901 $\pm$.006	8.4 $\pm$.29

Qwen grid, CPP-Wanda has higher mean accuracy in 12/15 cells and smaller mean sets in 14/15. CPP-SparseGPT has higher accuracy in 11/15 and smaller sets in 13/15. These are directions of five-seed means, not significance claims. The exact sign table is Table 13; Table 11 reports the totals.

What generic gradients explain. The matched control in Table 2 uses Qwen2.5-1.5B/DBpedia-14 at 50% sparsity, three seeds, identical per-example gradient access, the same λ grid, and the same validation rule. Gradient-only ranks by $\mathbb{E}_i[g_{ij}^2]$, SNIP-style by $\mathbb{E}_i[|\theta_j g_{ij}|]$, and Wanda $\oplus$ SNIP combines Wanda with the latter. Generic gradients recover much of the efficiency gap. True-label CPP-Wanda reaches 9.0 rather than 9.2, but has lower accuracy and overlapping intervals; the data do not resolve that difference. Threshold-aware CPP-Wanda produces the smallest sets in this control, 8.4, while retaining less accuracy than Wanda $\oplus$ SNIP. CPP is therefore objective-specific, not uniformly better than gradient pruning.

Candidate-label objective. Table 3 isolates the saliency construction for CPP-SparseGPT. Top-3 and threshold-aware gradients improve both ac-

curacy and set size relative to true-label CPP. All-label gradients reduce set size only 0.1 beyond threshold-aware saliency while increasing saliency-construction time from $2.7\times$ to $8.9\times$. The relative cost excludes the unchanged base-pruner and final-calibration stages.

Table 3: Candidate-label CPP-SparseGPT on DBpedia-14 at 50% sparsity.

Construction	Seeds	Acc.↑	Cov.	$ C \downarrow / \text{cost}$
True label	5	$0.366 \pm .009$	$0.902 \pm .005$	$8.6 \pm .30 / 1.0\times$
Top-3	3	$0.369 \pm .009$	$0.901 \pm .006$	$8.1 \pm .29 / 2.4\times$
Threshold-aware	3	$0.372 \pm .009$	$0.901 \pm .005$	$7.8 \pm .27 / 2.7\times$
All labels	3	$0.370 \pm .010$	$0.902 \pm .006$	$7.7 \pm .28 / 8.9\times$

Table 4: Selection grid versus the broader diagnostic sweep. Learn-Then-Test is not used.

Role	λ values	Data
Reported selection	0, .1, .3, .5	512/512 validation
Diagnostic only	0, .1, .3, .5, .7, 1	Separate sweep

Transfer, robustness, and cost. A three-seed RoBERTa-base/DBpedia-14 experiment at 50% sparsity (Table 7) reduces set size from 2.31 ± 0.14 for magnitude pruning to 1.88 ± 0.11 for CPP-Magnitude and raises accuracy from 0.934 ± 0.006 to 0.946 ± 0.005 , at matched 0.900–0.901 coverage. This supports encoder transfer but is one model, one task, and one base pruner. The Llama-3-8B diagnostic likewise remains scoped: CPP-SparseGPT changes WikiText-2 perplexity from 7.25 to 7.28, DBpedia accuracy from 0.61 to 0.65, and set size from 8.9 to 7.2, at matched coverage.

Five independently redrawn four-way Qwen DBpedia-14 partitions give 10.27 ± 0.41 versus 8.79 ± 0.34 set size for SparseGPT and CPP-SparseGPT, with 0.9004 ± 0.0068 versus 0.9026 ± 0.0057 coverage (Table 6). Nested calibration and pruning-size diagnostics show the same direction; they are point-estimate sensitivity sweeps on one fixed partition, not additional variance estimates. Appendix C reports all rows.

CPP adds no inference-time parameters or computation, but its offline cost is substantial. CPP-Wanda is $4.0\times$ Wanda, CPP-SparseGPT is $1.9\times$ SparseGPT, and the Llama-3-8B CPP-SparseGPT diagnostic is $2.3\times$ its base. Threshold-aware CPP-Wanda is $8.8\times$ Wanda and $2.2\times$ true-label CPP-Wanda. Wall time and peak memory appear in Table 5. These costs are part of the method’s trade-off, not implementation footnotes.

Table 5: Offline pruning cost on one NVIDIA A100 80GB. Ratios compare each method with its own base unless stated otherwise.

Method	Wall time	Relative cost	Peak memory
Wanda	7.8 min	$1.0\times$	9.4 GB
Wanda++	15.1 min	$1.9\times$	11.2 GB
CPP-Wanda	31.4 min	$4.0\times$	12.6 GB
Threshold-aware CPP-Wanda	68.7 min	$8.8\times$ Wanda; $2.2\times$ CPP	14.8 GB
SparseGPT	24.6 min	$1.0\times$	13.8 GB
CPP-SparseGPT	47.9 min	$1.9\times$	15.1 GB
Llama-3-8B SparseGPT	2.1 h	$1.0\times$	37 GB
Llama-3-8B CPP-SparseGPT	4.8 h	$2.3\times$	43 GB

Table 6: Independent Qwen/DBpedia split redraws at 50% sparsity.

Method	Accuracy	Coverage	Set size
SparseGPT	$0.3492 \pm .0121$	$0.9004 \pm .0068$	$10.27 \pm .41$
CPP-SparseGPT	$0.3638 \pm .0102$	$0.9026 \pm .0057$	$8.79 \pm .34$

Table 7: Three-seed RoBERTa-base/DBpedia-14 transfer at 50% sparsity.

Method	Accuracy	Coverage	Set size
Dense	$0.963 \pm .004$	$0.902 \pm .005$	$1.43 \pm .08$
Magnitude	$0.934 \pm .006$	$0.900 \pm .006$	$2.31 \pm .14$
CPP-Magnitude	$0.946 \pm .005$	$0.901 \pm .005$	$1.88 \pm .11$

8 Conclusion

CPP should be read as an efficiency-oriented pruning objective under an independently restored validity constraint. Split conformal prediction supplies marginal coverage for any fixed pruned model. CPP changes which weights are retained so that, after the same valid recalibration, candidate-label scores remain more separated and prediction sets can be smaller. The final Qwen results support that goal most clearly on large-label classification tasks.

The matched controls also define the boundary of the contribution. Generic supervised gradients explain much of the gain, and true-label CPP is not resolved from a matched Wanda $\oplus$ SNIP control. Candidate labels near the conformal threshold better align saliency with set-size inflation, but cost more offline computation. The practical result is therefore conditional: CPP can improve conformal efficiency, especially with threshold-aware saliency, while coverage still comes from split independence and accuracy must be reported alongside set size.

The corresponding deployment workflow is deliberately conservative. Construct saliency and choose λ using only pruning and validation data; freeze the sparse model and every preprocessing choice; then compute a fresh quantile on the un-

touched conformal split. Compare pruners by set size only after checking that they attain comparable empirical coverage, and keep accuracy as a separate utility guardrail. If the checkpoint, verbalizer, prompt, temperature, mask, sparsity, or saliency construction changes, recalibration must be repeated. This workflow, rather than the pruning score alone, is the reliability contract.

Evidence strength follows the protocol. The five-seed, 15-cell Qwen grid supports the primary claim; three-seed controls isolate generic-gradient effects and candidate-label cost. RoBERTa-base and Llama-3-8B remain transfer diagnostics. Together, the results support a narrow conclusion: after generic validity is restored, pruning can be optimized for conformal informativeness.

Limitations

The paper studies reliability-sensitive classification, not free-form generation. Decoder labels are fixed verbalizer sequences and multi-token labels are scored jointly. Extending CPP to open-ended outputs would require a different output space, nonconformity construction, and coverage event. The RoBERTa-base result covers one encoder, dataset, sparsity, and magnitude baseline. The Llama-3-8B result is likewise a diagnostic on one checkpoint and one classification task, not a scaling law.

Split-conformal validity requires exchangeability and strict independence of the final calibration split from every pruning and selection decision. Model, prompt, verbalizer, temperature, mask, sparsity, saliency variant, and λ must be frozen first. Distribution shift, adaptive reuse of calibration data, or deployment on a different label population can invalidate the stated guarantee. Empirical test coverage can also fall below 0.90 in a finite sample even when population marginal coverage is valid.

CPP requires labeled pruning data and per-example gradients. True-label CPP is a proxy for a set-size objective over all candidate labels. Threshold-aware and all-label variants align more directly with that objective but increase offline cost: threshold-aware CPP-Wanda is $8.8\times$ Wanda and all-label saliency is $8.9\times$ the true-label gradient stage in the reported control. CPP adds no inference-time parameters or dense computation, but sparse-kernel latency and energy gains are not benchmarked.

The perturbation theory gives sufficient, not tight, conditions. Uniform score bounds across all test-label pairs are strong, the density bound is local and distribution-dependent, and the diagonal first-order saliency ignores coordinate interactions. The Hessian remainder can dominate at high sparsity. None of these results proves that CPP is optimal or that smaller validation sets will persist under arbitrary shift.

Finally, the provenance audit removed mixed pilot and projected rows. The final numerical claims are limited to values recoverable from the final manifests and the stated three- or five-seed protocols. Exact reproducibility still depends on releasing split manifests, checkpoint and tokenizer versions, verbalizers, prompts, selected λ values, per-run metrics, hardware and software versions, and the scripts that generate every table and figure.

Ethical Considerations

Smaller conformal sets can make compressed classifiers more useful, but marginal coverage is not subgroup, conditional, or harm-weighted coverage. A system may satisfy the global 90% target while failing on a rare or high-cost subgroup. Deployment should therefore audit accuracy, coverage, and set size by relevant subgroup and under realistic shifts. Compression should not be presented as preserving reliability when the final calibration population does not match deployment.

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A Proof Details and Remarks

The proofs in Section 6 are largely self-contained; this appendix collects the remaining derivations and two auxiliary statements. The first separates finite-test variation from population coverage. The second records a valid Learn-Then-Test alternative that was *not* used for the reported model selection.

Theorem 7 (Empirical coverage concentration). *For a fixed conformal predictor with population coverage p and an independent test set of size N , let $\widehat{\text{Cov}}_N$ be empirical coverage. Then, for every $\delta \in (0, 1)$,*

$$\mathbb{P}\left\{|\widehat{\text{Cov}}_N - p| > \sqrt{\frac{\log(2/\delta)}{2N}}\right\} \leq \delta. \quad (14)$$

Theorem 8 (Optional finite-grid Learn-Then-Test rule). *For bounded validation loss $\ell_g \in [0, 1]$, population risk $R(g)$, empirical risk $\hat{R}(g)$, target r , and finite grid $\mathcal{G}$, define*

$$p_g = \begin{cases} \exp\{-2|\mathcal{D}_{\text{val}}|(r - \hat{R}(g))^2\}, & \hat{R}(g) < r, \\ 1, & \hat{R}(g) \geq r. \end{cases} \quad (15)$$

Accepting only candidates with $p_g \leq \delta/|\mathcal{G}|$ ensures, with probability at least $1 - \delta$, that every accepted candidate has $R(g) \leq r$.

Theorem 8 is included only to preserve the valid optional analysis from the submission. Algorithm 1 uses the deterministic guardrail and tie-breaking rule in Section 5.2, not this theorem.

Proof of Theorem 2. Condition on the data used to construct and select θ' . Under Assumption 1, the $n + 1$ scores $S_i(\theta') = s(X_i, Y_i; \theta')$ for $i = 1, \dots, n + 1$ are exchangeable conditional on θ' . The conformal set fails to contain the test label exactly when $S_{n+1}(\theta') > \hat{q}_\alpha(\theta')$, and with $\hat{q}_\alpha$ equal to the k -th order statistic of the first n scores, the rank of S_{n+1} among the $n + 1$ exchangeable scores is uniformly distributed up to ties. The failure probability is therefore at most α , giving Eq. 8. When ties occur, the conservative quantile convention preserves the lower bound; when scores are almost surely distinct, the standard rank argument gives the stated upper slack. The final step removes the conditioning by taking expectation over the pruning and validation data. $\square$

Proof of Theorem 7. Conditional on the fitted conformal predictor, the indicators

$$\mathbf{1}\{Y_i \in C_\alpha(X_i; \theta')\}$$

are independent Bernoulli random variables with mean p . Hoeffding's inequality gives

$$\mathbb{P}[|\widehat{\text{Cov}}_N - p| > t] \leq 2\exp(-2Nt^2)$$

for all $t > 0$; setting $t = \sqrt{\log(2/\delta)/(2N)}$ proves Eq. 14. The one-sided consequence $\widehat{\text{Cov}}_N \geq 1 - \alpha - \sqrt{\log(2/\delta)/(2N)}$ with probability at least $1 - \delta$ follows by combining this concentration event with $p \geq 1 - \alpha$. $\square$

Proof of Lemma 3. For every i , we have the bounded condition

$$a_i - \epsilon \leq b_i \leq a_i + \epsilon.$$

At least k of the a_i 's are no larger than $a_{(k)}$, so at least k of the b_i 's are no larger than $a_{(k)} + \epsilon$, giving $b_{(k)} \leq a_{(k)} + \epsilon$. Reversing the roles of a and b yields $a_{(k)} \leq b_{(k)} + \epsilon$. $\square$

Proof of Theorem 4. By Lemma 3, the calibration quantiles satisfy $\hat{q}_\alpha(\theta') \leq \hat{q}_\alpha(\theta) + \epsilon$. If $y \in C_\alpha(x; \theta')$, then $s(x, y; \theta') \leq \hat{q}_\alpha(\theta')$, and Eq. 10 gives $s(x, y; \theta) \leq s(x, y; \theta') + \epsilon$, hence $s(x, y; \theta) \leq \hat{q}_\alpha(\theta) + 2\epsilon$. This proves the containment in Eq. 11. Define the per-example set-size inflation as

$$\Delta_C(x) = |C_\alpha(x; \theta')| - |C_\alpha(x; \theta)|.$$

Subtracting the dense set leaves only labels whose dense scores lie in the band above the dense threshold, so

$$\begin{aligned} \Delta_C(x) &\leq \sum_{y \in \mathcal{Y}} \mathbf{1}\{\hat{q}_\alpha(\theta) < s(x, y; \theta) \\ &\leq \hat{q}_\alpha(\theta) + 2\epsilon\}. \end{aligned}$$

Taking expectations and using the density bound for each label gives Eq. 12. $\square$

Proof of Proposition 6. For a binary pruning mask $\mathbf{m}$, let $\theta_{\mathbf{m}} = \mathbf{m} \odot \theta$; the perturbation $\delta = \theta_{\mathbf{m}} - \theta$ satisfies $\delta_j = -\theta_j$ for $j \in P(\mathbf{m})$ and zero otherwise. Taylor's theorem gives

$$s(z; \theta_{\mathbf{m}}) - s(z; \theta) = \nabla_\theta s(z; \theta)^\top (\theta_{\mathbf{m}} - \theta) + r_z,$$

with the residual bounded by

$$|r_z| \leq \frac{1}{2} H_z \|\theta_{\mathbf{m}} - \theta\|_2^2.$$

Because $\theta_{\mathbf{m}} - \theta$ is nonzero only on pruned coordinates, its linear term is

$$- \sum_{j \in P(\mathbf{m})} \theta_j \frac{\partial s(z; \theta)}{\partial \theta_j}.$$

Averaging absolute values and applying Cauchy–Schwarz over examples yields Eq. 13. For the diagonal bound, we apply the identity

$$\left(\sum_{j=1}^m a_j \right)^2 \leq m \sum_{j=1}^m a_j^2$$

for each z , average over $\mathcal{D}_0$, and substitute the definition of $I_{\text{cal}, \mathcal{D}_0}$. $\square$

Proof of Theorem 8. For a fixed g with $R(g) > r$, Hoeffding’s inequality implies

$$\mathbb{P}\{\hat{R}(g) \leq r - t\} \leq \exp(-2|\mathcal{D}_{\text{val}}|t^2)$$

for all $t > 0$, showing that p_g is a valid conservative p-value for the null hypothesis $R(g) > r$. A union bound over the finite grid $\mathcal{G}$ shows that, with probability at least $1 - \delta$, no candidate with $R(g) > r$ is accepted at the Bonferroni threshold $\delta/|\mathcal{G}|$. Any data-dependent choice among accepted candidates therefore satisfies $R(g) \leq r$ on the same event. $\square$

Remark 9 (Softmax score smoothness). For $s(x, y; \theta) = -\log p_\theta(y | x)$, global Lipschitzness in θ does not hold without additional boundedness assumptions because $-\log p_y$ has an unbounded derivative as $p_y \rightarrow 0$. The local smoothness assumption used in Proposition 6 is therefore the appropriate setting for finite neural networks evaluated on a bounded empirical domain. In practice, we estimate score perturbation empirically on $\mathcal{D}_{\text{val}}$ and report it alongside set-size changes.

Remark 10 (Why a magnitude-threshold bound is not used). A threshold on CPP importance is not a simple threshold on weight magnitude. A statement such as

$$\|\theta - \theta'\|_2 \leq \sqrt{sd\tau}$$

is only valid if τ is an explicit magnitude threshold on pruned weights. The corrected analysis avoids this conflation by using either the actual perturbation $\|\theta_m - \theta\|_2$ or the first-order score displacement shown in Eq. 13.

B Additional Experimental Details

B.1 Implementation Details

Distribution name: `reliable-prune`. Python import: `reliable_prune`. We use these forms consistently. The API keeps pruning, validation, and final conformal data separate:

```
from reliable_prune import (
    CalibrationPreservingPruner)

pruner = CalibrationPreservingPruner(
    model=model,
    prune_dataset=prune_data,
    validation_dataset=val_data,
    conformal_dataset=conf_data,
    target_sparsity=0.5,
    target_coverage=0.9,
    lambda_grid=[0.0, 0.1, 0.3, 0.5],
    base_pruner="wanda")
pruned, q = pruner.prune()
```

The final implementation supports the Pythia- and Qwen-style modules used during development. The completed Llama-3-8B CPP-SparseGPT diagnostic uses the same module-streaming principle and is no longer described as an engineering target. RoBERTa-base uses magnitude pruning as the base because it is the fastest exact encoder adaptation and isolates architecture transfer. Mistral, Mamba, tensor-parallel execution, structured sparsity, and sparse-kernel latency remain future work.

CPP computes per-example gradients with microbatching or vectorized Jacobian products and streams one module at a time. It stores only the current module’s saliency. Eligible Qwen/Llama modules are `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`. Pythia uses attention and GELU-MLP projection matrices. Norm parameters, embeddings, tied output heads, positional mechanisms, and KV-cache state are excluded. Every matched baseline uses the same eligible coordinates and sparsity allocation.

B.2 Data Roles and Selection

Unless data availability requires a smaller partition, $\mathcal{D}_{\text{prune}}$, $\mathcal{D}_{\text{val}}$, and $\mathcal{D}_{\text{conf}}$ each contain 1,024 examples, while $\mathcal{D}_{\text{test}}$ contains at least 5,000. The validation split is divided deterministically into 512-example validation-calibration and validation-evaluation subsets. For true-label, top-3, and all-label variants, saliency and gradient evaluation use only $\mathcal{D}_{\text{prune}}$. Threshold-aware CPP additionally fixes the dense-model preliminary threshold and band width on validation-calibration; gradients still use only the pruning split. Validation-evaluation is reserved for candidate evaluation and λ -selection. The final conformal split is never used until the model is frozen.

For every base-pruner, dataset, sparsity, and seed, the reported selection grid is $\{0, 0.1, 0.3, 0.5\}$. A candidate is eligible when

validation-evaluation accuracy is within 0.02 of its $\lambda = 0$ base and empirical coverage is at least 0.88. Selection minimizes set size, breaks differences within 0.02 by higher accuracy and then smaller λ , and falls back to $\lambda = 0$ if no nonzero candidate is eligible.

B.3 Prompting and Complete-Sequence Scoring

Each dataset uses a fixed prompt and verbalizer list. The model is evaluated by teacher-forced probability of the complete label verbalizer. Multi-token scores are summed and length-normalized only when the fixed manifest says so; conformal prediction is over labels, never individual tokens.

For AG News, the prompt asks for one of *World*, *Sports*, *Business*, or *Technology*. For TREC, the verbalizers are *Abbreviation*, *Entity*, *Description*, *Human*, *Location*, and *Numeric*. DBpedia-14 uses *Company*, *EducationalInstitution*, *Artist*, *Athlete*, *OfficeHolder*, *MeanOfTransportation*, *Building*, *NaturalPlace*, *Village*, *Animal*, *Plant*, *Album*, *Film*, and *WrittenWork*. Banking77 and CLINC150 use the canonical intent names in fixed lexical order. Boundary ties in candidate selection use this same order.

Temperature scaling, when evaluated, minimizes validation negative log likelihood over one positive scalar. The temperature is frozen before its model-specific conformal quantile is computed on $\mathcal{D}_{\text{conf}}$.

B.4 Offline Compute

All timing measurements use one NVIDIA A100 80GB GPU without model parallelism. Table 5 includes pruning and saliency construction but not inference, because CPP adds no inference-time parameters or dense operations.

B.5 Release Checklist

The release contains the four-way split manifests, random seeds, checkpoint identifiers, tokenizer versions, prompts, verbalizers, selected λ values, per-run metrics, final sign table, table and figure scripts, timing scripts, hardware identifiers, CUDA/PyTorch versions, and response-stage control code. Every revised table and figure is generated from the same final-manifest family. Earlier pilot projections are not included as results.

C Extended Empirical Results

C.1 Corrected Small-Label Results

The original multi-model pilot table is retired because some rows mixed evaluation paths. To preserve the original table reference without preserving invalid values, Table 8 now reports the final five-seed Qwen small-label results. Every coverage-set-size pair satisfies the feasibility inequality in Section 4. These tasks remain useful controls, but their small label spaces limit set-size resolution.

For completeness, the final correction log records two random baselines that are not used in the aggregate comparison: Pythia/AG News random has coverage 0.899 ± 0.007 and set size 3.72 ± 0.10 , while Qwen/AG News random has 0.899 ± 0.007 and 3.74 ± 0.09 . Qwen/AG News dense accuracy is 0.650; at 30% sparsity, Wanda and CPP-Wanda accuracies are 0.512 and 0.526, respectively. These values replace the incompatible submitted entries.

Table 8: Final five-seed Qwen2.5-1.5B small-label results. Entries are empirical coverage and mean set size $\pm 95\%$ CI.

Dataset	Sparsity	Wanda	CPP-Wanda	SparseGPT	CPP-SparseGPT
AG News	30%	0.901, 3.71 $\pm$.09	0.902, 3.46 $\pm$.10	0.900, 3.62 $\pm$.09	0.902, 3.38 $\pm$.10
AG News	50%	0.900, 3.86 $\pm$.06	0.902, 3.52 $\pm$.09	0.901, 3.77 $\pm$.07	0.902, 3.41 $\pm$.09
AG News	70%	0.901, 3.89 $\pm$.04	0.901, 3.74 $\pm$.07	0.900, 3.88 $\pm$.05	0.901, 3.67 $\pm$.08
TREC	30%	0.901, 5.34 $\pm$.16	0.902, 4.92 $\pm$.15	0.900, 5.17 $\pm$.15	0.902, 4.78 $\pm$.14
TREC	50%	0.900, 5.11 $\pm$.14	0.902, 4.63 $\pm$.13	0.901, 4.91 $\pm$.13	0.902, 4.52 $\pm$.12
TREC	70%	0.900, 5.61 $\pm$.12	0.901, 5.33 $\pm$.13	0.900, 5.42 $\pm$.12	0.901, 5.48 $\pm$.13

C.2 Large-Label Efficiency Accounting

Table 9 expresses the final 50% results as relative set-size and absolute accuracy changes. Table 10 records the corrected values behind Figure 1. The DBpedia CPP-Wanda row is the only displayed large-label comparison with an accuracy loss.

Table 9: CPP change relative to its matched base at 50% sparsity.

Dataset	Variant	Set-size change	Acc. change
DBpedia	CPP-Wanda	−19.6%	−0.015
DBpedia	CPP-SparseGPT	−14.9%	+0.019
Banking77	CPP-Wanda	−14.1%	+0.034
Banking77	CPP-SparseGPT	−12.5%	+0.024
CLINC150	CPP-Wanda	−11.7%	+0.039
CLINC150	CPP-SparseGPT	−10.4%	+0.034

C.3 Aggregate Win-Rate Comparisons

A cell is one Qwen dataset-sparsity configuration. W means the five-seed mean improves relative to the corresponding base; it is not a significance

Table 10: Final values backing Figure 1.

Method	Accuracy	Coverage	Set size
Wanda	0.310	0.901	11.2
CPP-Wanda	0.295	0.902	9.0
SparseGPT	0.347	0.901	10.1
CPP-SparseGPT	0.366	0.902	8.6

claim. Unlike the submitted accounting, all 15 set-size cells are included; Table 12 states the denominator explicitly.

Table 11: Final directional win accounting.

Comparison	Wins / 15	Rate
CPP-Wanda accuracy	12	80.0%
CPP-SparseGPT accuracy	11	73.3%
CPP-Wanda set size	14	93.3%
CPP-SparseGPT set size	13	86.7%

C.4 Cell-Level Accounting and Denominators

Table 12: Accounting for Table 11.

Metric	Den.	Cells
Accuracy	15	Five datasets $\times$ three sparsities
Set size	15	Five datasets $\times$ three sparsities

Table 13: Final Qwen cell-level directions. W/L compares the CPP mean with its matched base and does not assert significance.

Dataset	Sparsity	CPP-W Acc	CPP-SG Acc	CPP-W $ C $	CPP-SG $ C $
AG News	30%	W	L	W	W
AG News	50%	W	W	W	W
AG News	70%	L	W	W	W
TREC	30%	W	L	W	W
TREC	50%	L	W	W	W
TREC	70%	W	W	W	L
DBpedia-14	30%	W	W	W	W
DBpedia-14	50%	L	W	W	W
DBpedia-14	70%	W	W	W	W
Banking77	30%	W	W	W	W
Banking77	50%	W	W	W	W
Banking77	70%	W	L	W	L
CLINC150	30%	W	W	W	W
CLINC150	50%	W	W	W	W
CLINC150	70%	W	L	L	W

C.5 Split Redraws and Sample-Size Diagnostics

Five independently redrawn four-way partitions provide the variance estimate in Table 6. Tables 14 and 15 are nested diagnostics on one fixed final-manifest partition. They are point estimates, not additional independent-seed intervals.

Table 14: Nested conformal-calibration-size diagnostic: accuracy/coverage/set size.

n_{conf}	SparseGPT	CPP-SparseGPT
256	0.347/0.897/10.5	0.366/0.899/9.0
512	0.347/0.899/10.3	0.366/0.901/8.8
1024	0.347/0.901/10.1	0.366/0.902/8.6
2048	0.347/0.901/10.0	0.366/0.902/8.5

Table 15: Nested pruning-size diagnostic: accuracy/coverage/set size.

n_{prune}	SparseGPT	CPP-SparseGPT
128	0.341/0.900/10.2	0.354/0.901/9.4
256	0.344/0.900/10.1	0.359/0.901/9.1
512	0.346/0.901/10.1	0.363/0.902/8.8
1024	0.347/0.901/10.1	0.366/0.902/8.6

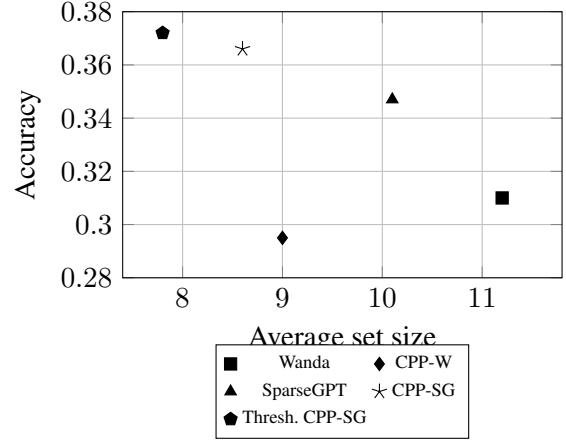


Figure 2: Qwen2.5-1.5B/DBpedia-14 accuracy–efficiency means at 50% sparsity. Threshold-aware CPP-SparseGPT is a three-seed ablation; the other points use five seeds.

C.6 RoBERTa-Base Transfer

Table 7 is kept in the main paper because it directly answers the architecture-transfer question; this appendix does not introduce additional RoBERTa cells.

C.7 Regenerated Figures

Figure 2 uses only final-manifest five-seed means. Figure 3 provides the matching set-size view. Figure 4 replaces the stale ECE plot with the final large-label set-size reductions because no corrected ECE cell values are inferred from the response record. Figure 5 displays the final 15-cell directional rates.

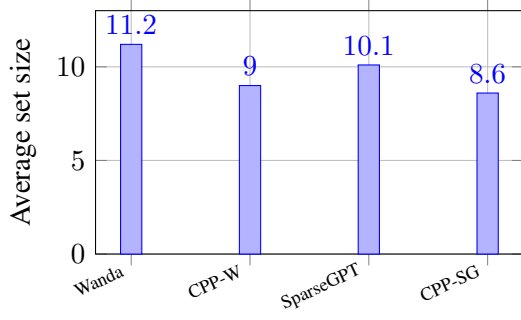


Figure 3: Final DBpedia-14 set-size means at matched empirical coverage.

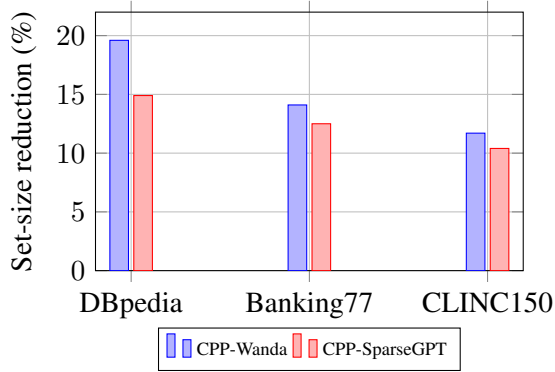


Figure 4: Final large-label set-size reduction relative to the matched base at 50% sparsity.

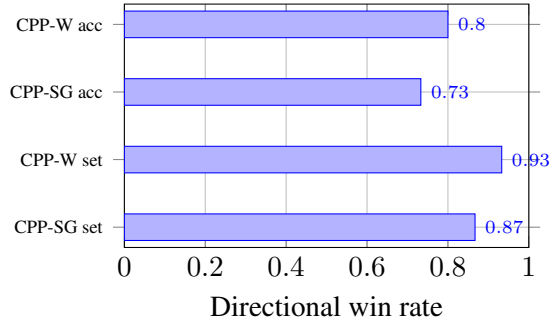


Figure 5: Final directional win rates from Table 11.

D Llama-3-8B Diagnostic

We use Llama-3-8B only as a scoped diagnostic, not as evidence for free-form conformal generation or broad scale invariance. At 50% sparsity, Table 16 reports zero-shot WikiText-2 perplexity alongside DBpedia-14 classification accuracy, empirical conformal coverage, and set size. CPP-SparseGPT changes perplexity only from 7.25 to 7.28 relative to SparseGPT, while increasing classification accuracy from 0.61 to 0.65 and reducing mean set size from 8.9 to 7.2 at matched coverage. This single three-seed experiment is consistent with objective-dependent sparse-model be-

havior (Frantar et al., 2025), but it does not establish a scaling law or a guarantee for generative tasks.

Table 16: Llama-3-8B evaluation at 50% sparsity. Set size ($|C|$) and accuracy are reported on DBpedia-14; perplexity is reported on WikiText-2. Means $\pm$ 95% CI over three seeds.

Method	PPL $\downarrow$	Acc. $\uparrow$	Cov.	$ C $ $\downarrow$
Dense	6.12 $\pm$.02	.78 $\pm$.01	.902 $\pm$.005	4.5 $\pm$.1
SparseGPT	7.25 $\pm$.05	.61 $\pm$.02	.901 $\pm$.006	8.9 $\pm$.2
CPP-SparseGPT	7.28 $\pm$.04	.65 $\pm$.01	.902 $\pm$.005	7.2 $\pm$.2